High-Order Winslow smoothing/untangling

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Codes live here.

0 The 2016 formulation (brief recap)

The Winslow equations in physical coordinates are

$$g^{ij}\,\partial_i\partial_j x_k = 0, \qquad k = 1, \ldots, n, \tag{1}$$

where g^{ij} are the contravariant metric components of the (nonlinear) mapping from the computational domain C to the physical domain D. The nonconservative form does not discretize cleanly in a CG framework, so the 2016 paper introduces auxiliary variables $\alpha_j = -\partial_i(g^{ij})$ and recasts the system in split form:

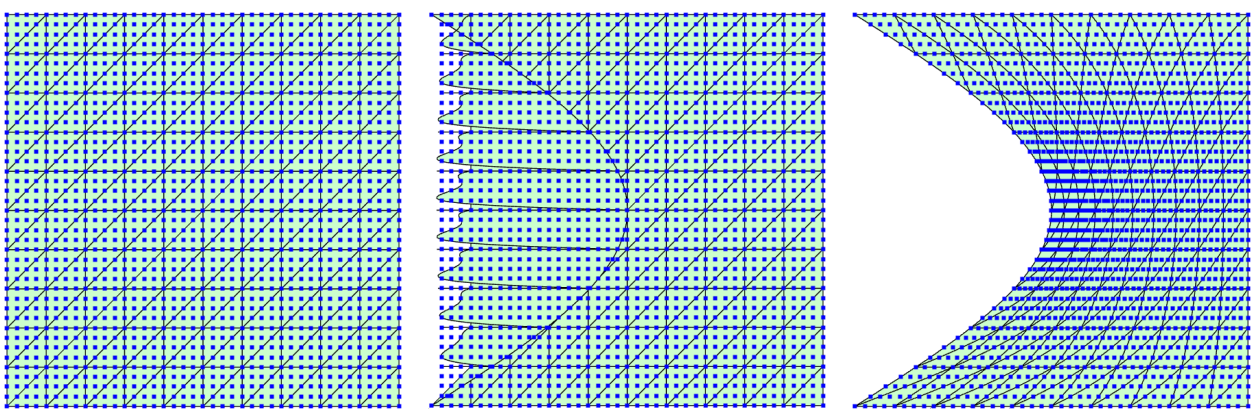
$$\partial_i(g^{ij}) + \alpha_j = 0, \tag{2}$$

$$\partial_i(g^{ij}\,\partial_j x_k) + \alpha_j\,\partial_j x_k = 0, \tag{3}$$

with Dirichlet data $\boldsymbol{x} = \boldsymbol{x}_{\text{bnd}}$ on ∂C. The point of this split form is to 1) mimic the what finite difference Winslow people are doing and 2) enable Picard iteration, which is the "magic" here.

0.1 Numerical Results/Tests

The code `w2.m` does the following in Figure 1.



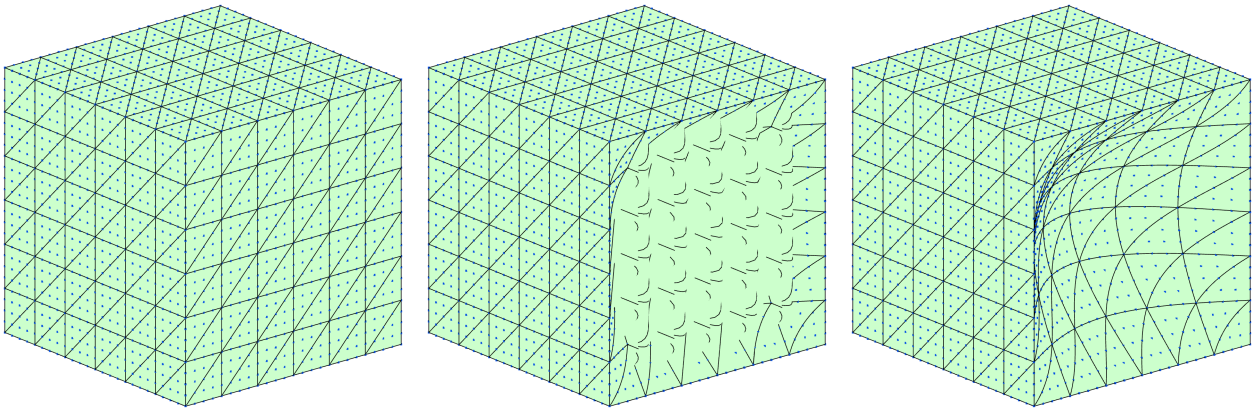
(a) Reference mesh. (b) Initial configuration. (c) After Winslow

Figure 1: (a) Structured 11×11 reference mesh at $p = 4$. (b) Initial configuration: interior nodes unchanged, boundary on side 4 displaced by $0.5\sin(\pi y)$, producing heavy tangling. (c) After Winslow iterations, interior nodes have redistributed to absorb the boundary deformation; all elements valid.

It is straightforward to ask Claude Code to write the 3D analogue `w2_3d.m` in Figure 2. To assess mesh quality we use the scaled Jacobian of Fortunato–Persson [1],

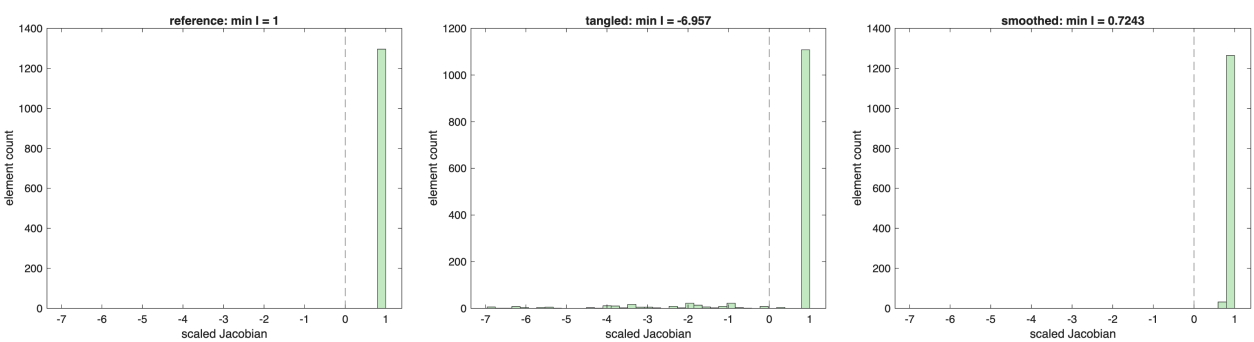
$$I \;=\; \frac{\min_{\xi\in\kappa} J(\xi)}{\max_{\xi\in\kappa} J(\xi)}, \qquad I \le 1, \tag{4}$$

approximated at the Gauss quadrature points of each element κ. For a straight-sided simplex, J is constant across κ and $I = 1$ exactly; $I \le 0$ indicates a degenerate or inverted element. Histograms for `w2_3d.m` are given in Figure 3.



(a) Reference mesh. (b) Initial configuration. (c) After Winslow

Figure 2: (a) Structured $7\times7\times7$ tetrahedral reference mesh of the unit cube at $p = 4$. (b) Initial configuration: interior nodes unchanged, boundary on side 1 ($x = 0$) displaced by $0.4\sin(\pi y)\sin(\pi z)$, producing heavy tangling. (c) After Winslow iterations, interior nodes have redistributed to absorb the boundary deformation; all elements valid.



(a) Reference (b) Tangled (c) Smoothed

Figure 3: Per-element scaled Jacobian $I = \min_\xi J(\xi)/\max_\xi J(\xi)$ (Fortunato–Persson 2016, Sec. 4) for each configuration. (a) Reference: all elements straight-sided simplex, $I = 1$. (b) Tangled: long negative tail with $\min I = -6.96$; boundary-adjacent elements inverted. (c) Smoothed: all $I > 0$, most near 1, with $\min I \approx 0.70$ on elements bordering the indentation.

1 W modification to achieve reference element shape

The 2016 formulation produces a physical mesh whose element shapes are inherited from the reference mesh: the contravariant metric g^{ij} in the split-form system of Section 0 is built from the reference element geometry, so a skewed reference produces skewed physical elements. The W-modification below decouples the target shape from the reference element shape.

1.1 Target metric

For each reference element $K \in \mathcal{T}_h$, let $W_K \in \mathbb{R}^{n\times n}$ denote the Jacobian of an affine map from an ideal element (e.g. a unit equilateral simplex or unit square) to the reference element geometry. The total map from the ideal element to the physical element then has Jacobian $J_{\text{tot}} = JW_K^{-1}$, where $J = \partial x/\partial \xi$ is the reference-to-physical Jacobian.

The covariant and contravariant metrics of the total map are

$$\tilde{g}_{ij} = (J_{\text{tot}}^T J_{\text{tot}})_{ij} = (W_K^{-T} g W_K^{-1})_{ij}, \qquad \tilde{g}^{ij} = (W_K g^{-1} W_K^T)_{ij} = W_{K,ik}\, g^{kl}\, W_{K,jl}. \tag{5}$$

With $W_K = I$ for all K, $\tilde{g}^{ij} = g^{ij}$ and the 2016 formulation is recovered.

1.2 Modified split form

We replace g^{ij} by $\tilde{g}^{ij}$ in the split-form system of Section 0, introducing a modified auxiliary variable $\tilde{\alpha}_j = -\partial_i \tilde{g}^{ij}$:

$$\partial_i(\tilde{g}^{ij}) + \tilde{\alpha}_j = 0, \tag{6}$$

$$\partial_i(\tilde{g}^{ij}\, \partial_j x_k) + \tilde{\alpha}_j\, \partial_j x_k = 0, \tag{7}$$

with Dirichlet data $x = x_{\text{bnd}}$ on ∂C. The equivalence with the nonconservative form $\tilde{g}^{ij}\partial_i\partial_j x_k = 0$ follows from (6) by the same product-rule argument as in the unmodified case.

Heuristic nature of the extension. The classical Winslow system $g^{ij}\partial_i\partial_j x_k = 0$ is the reformulation, via change of variables, of $\Delta_x \xi_k = 0$—the Euler–Lagrange equation of the Dirichlet energy $\int_D |\nabla_x \xi|^2\, dx$ for the inverse map. The substitution $g \mapsto \tilde{g}$ in (6)–(7) is not, to the authors' knowledge, the Euler–Lagrange equation of any natural energy functional in x. In particular, for piecewise-constant W_K the map $\xi \mapsto W_K\xi$ is a per-element linear reparameterization of the reference coordinates, and harmonic maps are invariant under such reparameterizations, so the natural variational interpretation collapses to the unmodified Winslow system. We therefore regard (6)–(7) as a heuristic extension of the 2016 numerical scheme, parameterized so as to retain its Picard structure, rather than as the discretization of a derived PDE. Consequences: no Radó–Kneser–Choquet analog applies, and convergence of the resulting Picard iteration is assessed empirically.

1.3 Continuity of W across element interfaces

A piecewise-constant W_K renders $\tilde{g}^{ij}$ discontinuous across element interfaces. This is not a new obstruction: g^{ij} itself is already discontinuous in the 2016 formulation, since the CG space V_h^p yields $x_h \in C^0$ but $\nabla x_h \notin C^0$. The weak projection defining α_h, which includes only external boundary fluxes, smooths these jumps into V_h^p through the CG mass matrix.

The same mechanism handles jumps in $\tilde{g}^{ij}$, provided the amplitude of the W_K-induced jumps remains comparable to those already present in g^{ij}. When the reference mesh contains neighboring

elements of strongly differing shape, we avoid amplifying interface jumps by replacing W_K with a continuous nodal field $W_h \in V_h^p$, constructed by lumped L^2 projection: at each node $\mathbf{x}_n$,

$$W_h(\mathbf{x}_n) = \frac{\sum_{K \ni \mathbf{x}_n} |K|\, W_K}{\sum_{K \ni \mathbf{x}_n} |K|}, \tag{8}$$

with W_h then interpolated across element interiors via the standard shape functions. This corresponds to L^2 projection with a lumped mass matrix and is computed once per run. The resulting $\tilde{g}^{ij}$ inherits the same regularity class as the unmodified g^{ij}, and the $\tilde{\alpha}$ projection is unchanged at the implementation level.

1.4 Boundary treatment

At boundary nodes adjacent to curved geometry, an isotropic target W_h can conflict with geometric constraints. Where this occurs we align one principal direction of W_h tangent to the boundary, leaving the magnitude prescribed by the local sizing. Details are deferred to the implementation section.

1.5 Effect on the Picard iteration

The assembly structure of the 2016 Picard iteration is preserved: the CG mass matrix in the $\tilde{\alpha}$ solve is unchanged, and the stiffness matrix in the x solve is assembled from $\tilde{g}^{ij}$ and $\tilde{\alpha}_j$ via the same routines as the unmodified case. The convergence rate, however, depends on spectral properties of the iteration operator, which change with W. We report iteration counts empirically in the results section.

1.6 Numerical Results/Tests

In order to test the effectiveness of this W:

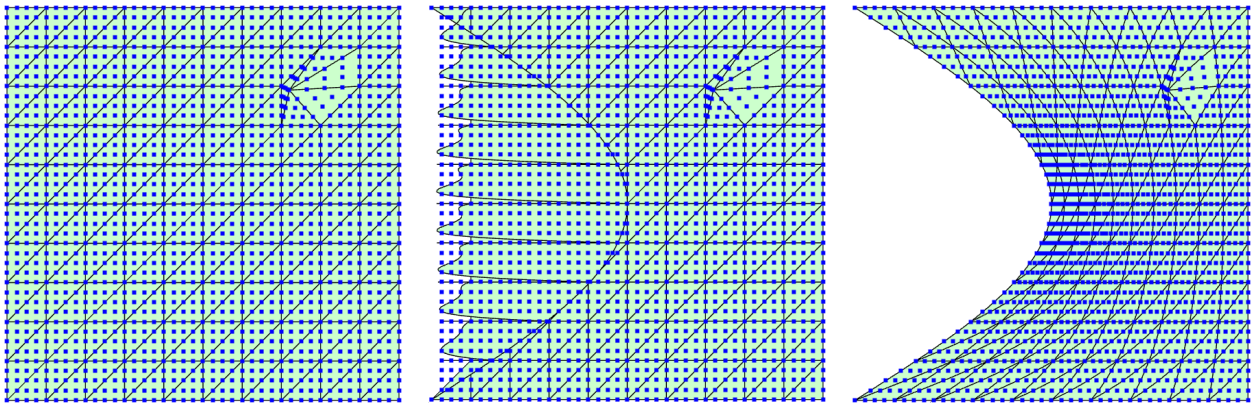
- Repeat the experiments `w2.m` and `w2_3d.m` with a skewed but valid reference mesh of the unit/cube. Observe that the current Persson-Fortunato setting will behave correspondingly according to the reference mesh.
- Implement W modification. Check that this makes the physical mesh truly agnostic to the quality of the skewed (but for now, still valid/untangled) reference mesh.

`w2.m` is reproduced but with some poorly shaped elements in the reference mesh. The code that does this is `w2_skewed.m`. The results are in Figure 4.

The scaled Jacobian (4) is blind to this regime: any simplex with straight edges has $I = 1$ regardless of aspect ratio. To capture straight-sided shape distortion we additionally report the Liu–Joe inverse mean-ratio shape distortion metric [2, 3],

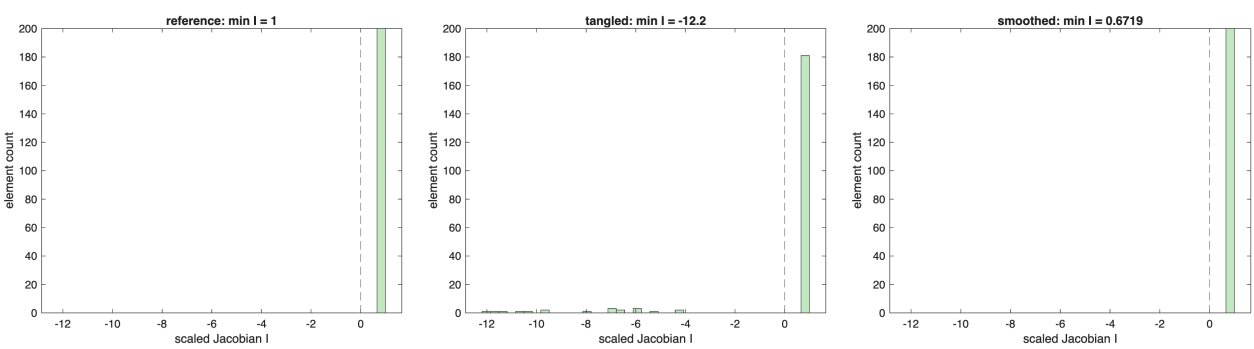
$$\eta \;=\; \frac{\|\boldsymbol{J}\|_F^2}{d\,|\det \boldsymbol{J}|^{2/d}}, \qquad \eta \in [1, \infty), \tag{9}$$

with $\eta = 1$ for a conformal (ideal) element and $\eta \to \infty$ as the element degenerates. The scaled Jacobian distribution is given in Figure 5 and the shape distortion distribution in Figure 6.



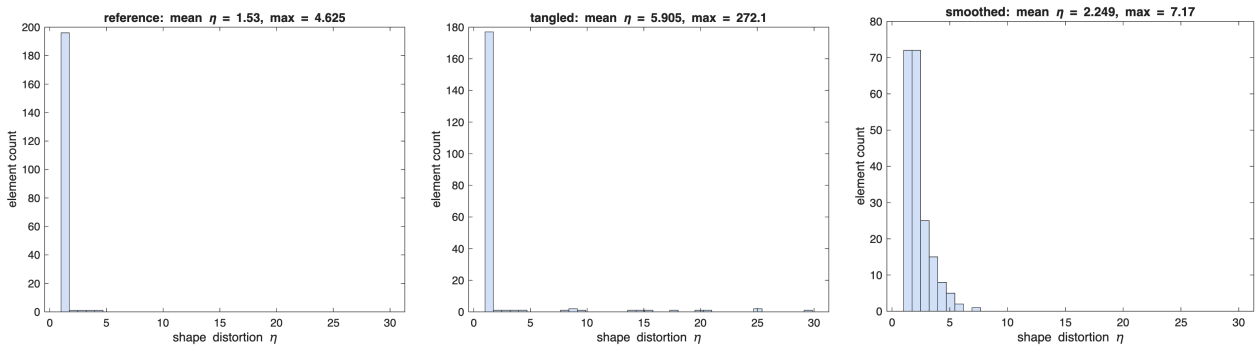
(a) Reference mesh. (b) Initial configuration. (c) After Winslow

Figure 4: Baseline test: structured 11×11 mesh at $p = 4$ with one interior vertex manually displaced toward a neighbor to create a single sliver cluster in the upper-right. (a) Skewed reference with $\max \eta = 4.63$. (b) Boundary on side 4 displaced by $0.5 \sin(\pi y)$, producing heavy tangling near the perturbed edge on top of the reference-induced sliver. (c) After unmodified Winslow: elements valid, but reference skew persists, with $\max \eta = 7.17$ on the sliver in the upper-right.



(a) Reference (b) Tangled (c) Smoothed

Figure 5: Per-element scaled Jacobian $I = \min_\xi J(\xi) / \max_\xi J(\xi)$ (Fortunato–Persson 2016, Sec. 4). (a) Reference: all elements straight-sided simplex, so $I = 1$ identically—the metric is blind to the sliver clusters present in the mesh. (b) Tangled: long negative tail, $\min I = -12.2$. (c) Smoothed: all $I > 0$, $\min I \approx 0.67$ on elements bordering the indentation.



(a) Reference (b) Tangled (c) Smoothed

Figure 6: Per-element shape distortion metric $\eta = \|J\|_F^2/(d\,|\det J|^{2/d})$ [2, 3] for each configuration. $\eta \in [1, \infty)$ with $\eta = 1$ at conformal (ideal) elements; unlike I, η detects straight-sided slivers. (a) Reference: $\max \eta = 4.63$, mean 1.53—the right tail corresponds to the sliver cluster around the moved vertex. (b) Tangled: $\max \eta = 272$. (c) Smoothed: $\max \eta = 7.17$, mean 2.25—the sliver cluster persists as the worst element in the smoothed output, motivating the W-modification in Section 1.

1.7 Numerical experiments with the W-modification (negative result)

We implemented the split-form modification of Section 1—literally substituting $\tilde{g}^{ij} = W_K g^{-1} W_K^T$ in place of g^{ij} in both `sm1alpha2d` and `sm12d` of the Picard solver—and ran it on the single-sliver test of Figure 4. The implementation lives in `elliptic_smoothing_W.m`, with the per-element W_K built by `compute_WK_simplex.m` from an ideal equilateral triangle. A regression test (`demos/w_regression.m`) verifies that the modified solver reproduces the unmodified output bit-for-bit when $W_K = I$ at every element, so the wiring is trustworthy.

The experimental conclusion is negative. The W-modification converges in most of the variants we tried, but none of them produce a smoothed mesh that is meaningfully better than the unmodified Winslow output on the sliver test. Below we report the specific variants we tried and why each fails.

Piecewise-constant W_K. The simplest implementation: feed the raw per-element W_K directly into the substituted split form, with no smoothing across element interfaces. For the uniform unskewed reference mesh this already diverges: adjacent triangles in each cell of `mshsquare` have different corner orderings, so W_K jumps by a rotation/reflection at every interior edge. The jumps in $\tilde{g}^{ij}$ that result are too large for the weak $\tilde{\alpha}$-projection to absorb, and the Picard iteration blows up. Restricting $W_K \neq I$ to only the four sliver-cluster elements, with $W_K = I$ everywhere else, also diverges—even four interfaces with a non-trivial jump are enough to destabilize.

Nodal projection onto P^1 (linear corners only). As suggested in Section 1.3, we project W_K to a continuous field W_h by a lumped L^2 projection onto the linear corner vertices of the mesh (Eq. (8)), then interpolate linearly to all DG nodes for use at Gauss points. This stabilizes the Picard iteration but also washes the correction out. On the single- sliver test, the sliver reference vertex at $\xi = (0.72, 0.79)$ is mapped by the smoothed output to: $(0.7964, 0.79)$ under unmodified Winslow and $(0.8029, 0.79)$ under W-modified (P^1-projected) Winslow. The target (fully un-skewed) is $(0.8, 0.8)$. The unmodified solver already performs 96% of the x-correction on its own by virtue of being a harmonic smoother; the W-modification contributes an additional 0.006

shift, within the noise floor of the iteration. Neither method moves the y-coordinate at all, because the boundary perturbation in `w2.m` is purely in x and the Winslow system decouples component-wise. Globally the W-modified output is in fact *worse* than the unmodified one: $\max \eta = 19.2$ versus 7.17, with the worst element having migrated from the sliver location to the lower-middle of the domain.

Nodal projection onto V_h^p (all CG nodes). The notes' derivation (Section 1.3) places $W_h \in V_h^p$, i.e., on the full order-p CG space, not just the linear corners. We implemented this variant as well. On the uniform unskewed mesh (`w2.m`) it converges but produces a smoothed mesh with $\min J = -2.6 \times 10^{-4}$, i.e. *inverted* elements, on a problem where the unmodified solver has no difficulty. On the sliver test the iteration fails to converge altogether. Full V_h^p preserves more of the per-element W_K near element interiors and less of the corner averaging, and this extra expressiveness is enough to push the heuristic past stability.

Weighting schemes in the lumped projection. Equation (8) uses the element area $|K|$ as the lumping weight. This down-weights slivers (which have small $|K|$) relative to their regular-sized neighbors, so the sliver's W_K is largely drowned out in the projected W_h. We replaced $|K|$ by η_K, $\eta_K - 1$, and η_K^2 in an attempt to boost the sliver's contribution, but the sliver vertex image remained within ± 0.01 of the unmodified result across all schemes, and global $\max \eta$ stayed between 19 and 22. The weighting choice does not meaningfully move the numerical outcome.

Sliver-only W. Restricting $W_K \neq I$ to only the four sliver-cluster elements (identified by an η threshold on the reference) and projecting only those via P^1 gives a sliver image of $(0.7966, 0.79)$—indistinguishable from the unmodified $(0.7964, 0.79)$. The averaging with the identity-valued W_K of the surrounding elements at shared CG corners neutralizes the sliver correction almost entirely. The piecewise-constant and V_h^p variants of this experiment both diverge.

Summary. Table 1 collects the results. Across every variant we tried—three projection choices (piecewise constant, P^1, V_h^p), four lumping weights (area, η, $\eta - 1$, η^2), and a sliver-only mask—the W-modification either diverges, or converges to an output whose sliver image agrees with the unmodified Winslow result to within the iteration's noise floor. The reason is visible by inspection: the unmodified Winslow solver, being harmonic, already pulls the interior sliver vertex 96% of the way back to the un-skewed target. Any projection that is stable enough to keep Picard converging is also one that averages the local W_K correction heavily with its identity-valued neighbors, leaving only a few-tenth-of-a-percent shift that cannot close the remaining 4% gap.

Interpretation. This is consistent with the caveat already stated in Section 1.2: the modified split form is not the Euler–Lagrange equation of any natural energy, and there is no discrete minimum principle guaranteeing that the Picard fixed point is a meaningful "W-harmonic" map. Empirically, for interior sliver defects on an otherwise well-shaped mesh, the W-modification as written in Section 1 does not deliver the reference-agnostic behaviour that the classical formulation lacks—it just reshuffles the distortion without reducing it. Whether the construction can be rescued by machinery outside the scope of Section 1 (for instance an anisotropic spatially-varying ideal, damped/continuation Picard, or a proper variational reformulation with an H^1-type energy in x) is an open question.

Table 1: Sliver reference vertex $\xi = (0.72, 0.79)$ image under different W-modification variants, on the single-sliver test of Figure 4. "Target" denotes the un-skewed position $(0.8, 0.8)$. Unmodified Winslow is shown for reference.

variant	sliver image	global $\max \eta$	notes
unmodified Winslow	$(0.7964, 0.79)$	7.17	baseline
piecewise-const W_K	—	—	diverges
P^1 projection, area weight	$(0.8029, 0.79)$	19.2	converges
P^1 projection, η weight	$(0.8010, 0.79)$	19.5	converges
P^1 projection, η^2 weight	$(0.7937, 0.79)$	21.7	converges
P^1 projection, $\eta - 1$ weight	—	—	diverges
V_h^p projection	—	—	diverges or $\min J < 0$
sliver-only W, P^1	$(0.7966, 0.79)$	7.05	$\approx$ unmodified
sliver-only W, piecewise const	—	—	diverges
sliver-only W, V_h^p	—	—	diverges
target (un-skewed)	$(0.80, 0.80)$	—	

References

[1] Meire Fortunato and Per-Olof Persson. High-order unstructured curved mesh generation using the Winslow equations. *Journal of Computational Physics*, 307:1–14, 2016.

[2] Anwei Liu and Barry Joe. Relationship between tetrahedron shape measures. *BIT Numerical Mathematics*, 34(2):268–287, 1994.

[3] A. Shi and P.-O. Persson. Local element operations for curved simplex meshes. *International Journal for Numerical Methods in Engineering*, 2023. Author manuscript.